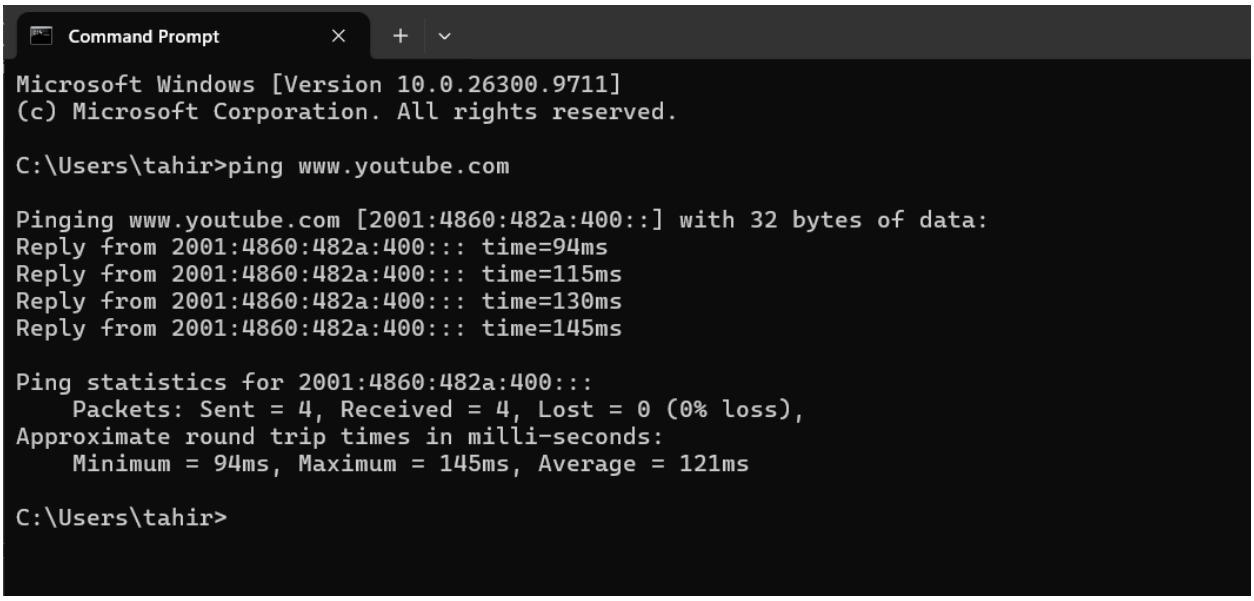
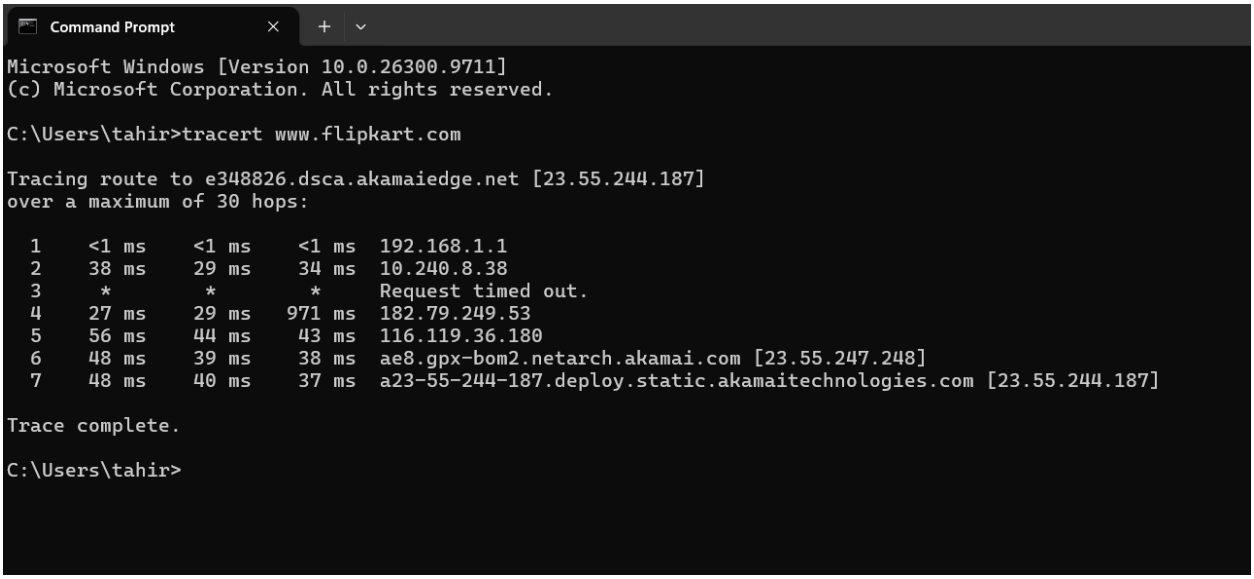
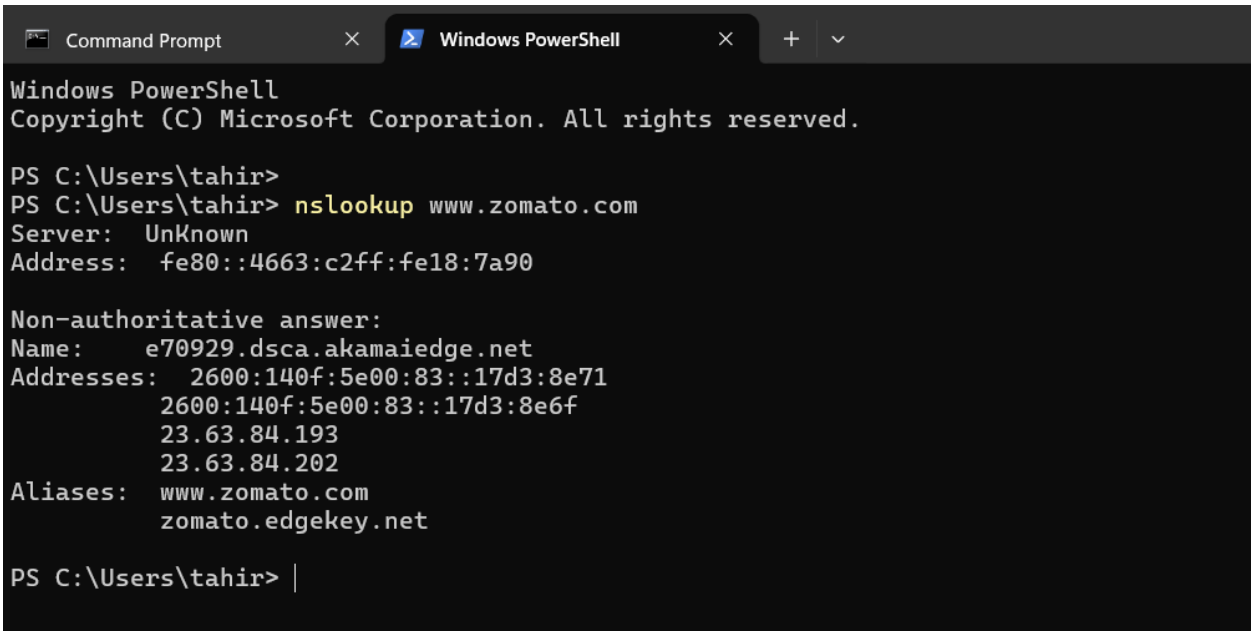
1.Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.



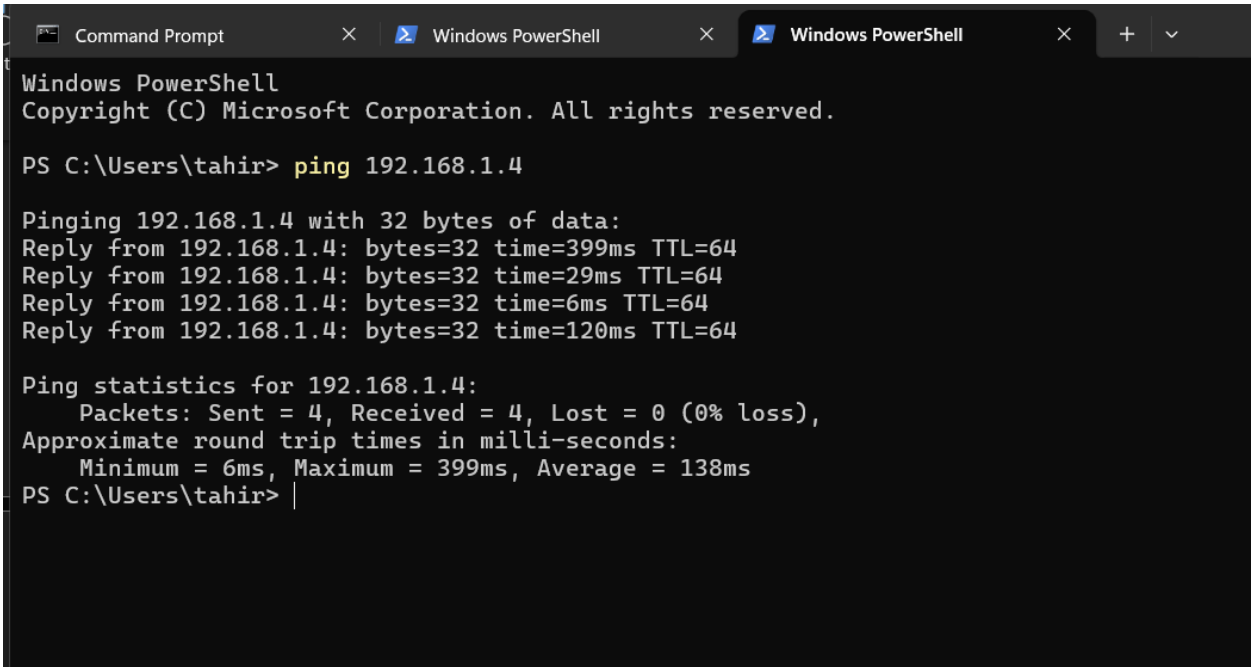
2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint: Each hop represents a network device your data passes through to reach the destination.



3.Use the nslookup tool to find the IP address of www.zomato.com and record the result.



4.Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.



If the device is offline, the ping command will show “Request timed out” because the device is not responding to the ping request.

5.Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint: Give a specific example for each tool based on your own test results above.

1. **Website not loading**: I use ping to check if the website is responding.
2. **Slow connection**: I use traceroute to find where the connection is slow.
3. **DNS issues**: I use nslookup to check if a website name, like google.com, is getting the correct IP address.
4. When I ping an offline device, I may get “Request timed out.”

These three tools help me find and understand network problems easily.